

Noah Rover End-User Manual

RC Controller Behaviors, Mission Flow, Obstacle Avoidance, Edge Guidance, Docking, and LCD Meanings

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1) Safety First

- Keep people and pets clear before arming, undocking, docking, or mission start.
- Use RC mode (CH11 high) as your operator override for mission and obstacle-avoidance behaviors.
- If an unexpected sequence is active, use dock switch OFF to stop dock/undock sequences and hold the rover.
- On any abnormal behavior, stop movement first, then troubleshoot.

2) RC Switch/Stick Map

Control	Channel	What It Does
Steering	CH1	Manual steering in RC mode.
Throttle	CH3	Manual forward/reverse speed in RC mode.
Yaw / Spin	CH4	Spin-in-place behavior in RC mode when outside center.
Claw Manual Enable	CH5	Enable/disable claw manual controls.
Dock / Undock	CH7	High = undock sequence. Low = dock authorization during mission, test-dock in RC/manual.
Actuator	CH8	Claw actuator command when claw manual is enabled.
Obstacle Avoidance Toggle	CH9	High disables lidar avoidance, low enables it.
RC / Mission Mode	CH11	High = RC mode. Low = mission request/start.
Belt	CH12	Belt command when claw manual is enabled.

3) Mission Mode Behavior (CH11)

- Mission request/start: CH11 low after a valid high-to-low switch transition.
- RC mode: CH11 high disables mission mode and returns manual drive control.
- If mission is requested during undock, mission start is queued until undock completes.
- At mission start, avoidance state is reset and a short startup grace window is applied.

4) Undock Behavior (CH7 high)

- CH7 high starts undock sequence only. It does not directly start mission.
- Undock uses pitch transitions to detect ramp travel and drives clear of dock before stopping.
- If mission was requested during undock (CH11 low), mission auto-starts right after undock complete.

5) Docking Behavior on Return

- After delivery, rover returns to the recorded undock latitude/longitude and aligns to undock heading.
- If dock/undock switch was NOT turned off during mission, rover proceeds to automatic light-seeking docking.
- If CH7 was switched OFF during mission, rover waits for an explicit dock command (CH7 low) before docking.
- When dock command is received while mission is active, docking is authorized for the return handoff.

6) Object Avoidance: What Triggers It

- Obstacle avoidance is active only when mission mode is on, RC link is valid, and avoidance toggle is enabled (CH9 low).
- Primary trigger is a blocked front arc: LiDAR front zones (11/12) red, or RealSense front vision zones red when fresh.
- A short block-confirm delay is used to reduce false triggers from momentary noise.
- Avoidance turns toward nearest non-front green zone, then creeps forward until corridor is stable-clear.
- First-leg commitment gate: at mission start, stale pre-start detections are ignored until rover commits to the first leg.

7) Edge Detection / RealSense Guidance

- RealSense edge guidance assists steering during mission two-wheel forward segments.
- Guidance favors visible sidewalk edge geometry with confidence checks and stale-data timeout.
- If edge confidence drops, steering authority latches then fades back to GPS guidance.
- RealSense obstacle persistence can trigger fallback behavior if path stays blocked too long.

8) LCD Screen Meanings

Noah uses three LCD screens:

- LCD 1 (mouth): animated smile/frown. Smile means all-systems-good; frown indicates at least one major subsystem not connected.
- LCD 2 (status): device connections, battery voltage, visible satellites, and heading.
- LCD 3 (vision): left/right edge seen and confidence, edge offsets, avoidance state, LiDAR front state, RealSense front state, and IRLock visibility.

LCD 3 shorthand:

- AV: CLR = clear, BLK = currently blocked, TMO = avoidance timeout.
- L: G/Y/R = LiDAR front-arc status green/yellow/red.
- V: G/Y/R = RealSense front object zone status green/yellow/red.
- IRL: 1/0 = IRLock light fresh/not fresh.

9) Typical End-User Sequences

Start mission with undock:

- Arm rover.
- Set CH7 high to undock.
- Set CH11 low during undock or after undock to start mission.

Require manual dock permission on return:

- During mission, flip CH7 to OFF once.
- Rover returns and aligns, then waits.
- Flip CH7 to dock/low to authorize and begin docking.

10) Troubleshooting Quick Notes

- Mission will not start: verify RC channels are valid and CH11 has a proper high-to-low transition.
- Rover waiting near dock and not docking: likely manual dock is required; set CH7 to dock/low.
- Avoidance not reacting: ensure mission mode is active and CH9 is set to enable avoidance.
- Unexpected early avoidance at mission start should be reduced by first-leg commitment logic; if seen, re-check GPS readiness and sensor freshness.

End of Manual